Accidents are hard to avoid because of sudden, unpredictable developments. Scientists, inventors, and experts often significantly underestimate the time it takes for a groundbreaking technological advancement to become a reality. The Wright brothers famously claimed that powered flight was fifty years away, just two years before they achieved it. Lord Rutherford, a prominent physicist and the father of nuclear physics, dismissed the idea of extracting energy from nuclear fission as "moonshine," only for Leo Szilard to invent the nuclear chain reaction less than 24 hours later. Similarly, Enrico Fermi expressed 90 percent confidence in 1939 that it was impossible to use uranium to sustain a fission chain reaction—yet, just four years later he was personally overseeing the first reactor [81].

AI development could catch us off guard too. In fact, it often does. The defeat of Lee Sedol by AlphaGo in 2016 came as a surprise to many experts, as it was widely believed that achieving such a feat would still require many more years of development. More recently, large language models such as GPT-4 have demonstrated spontaneously emergent capabilities [82]. On existing tasks, their performance is hard to predict in advance, often jumping up without warning as more resources are dedicated to training them. Furthermore, they often exhibit astonishing new abilities that no one had previously anticipated, such as the capacity for multi-step reasoning and learning on-the-fly, even though they were not deliberately taught these skills. This rapid and unpredictable evolution of AI capabilities presents a significant challenge for preventing accidents. After all, it is difficult to control something if we don't even know what it can do or how far it may exceed our expectations.

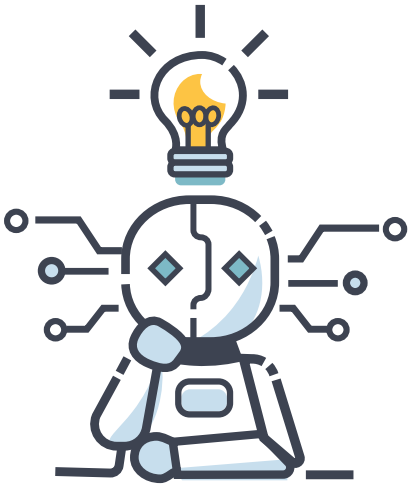

Figure 12: New capabilities can emerge quickly and unpredictably during training, such that dangerous milestones may be crossed without our immediate knowledge.

It often takes years to discover severe flaws or risks. History is replete with examples of substances or technologies initially thought safe, only for their unintended flaws or risks to be discovered years, if not decades, later. For example, lead was widely used in products like paint and gasoline until its neurotoxic effects came to light [83]. Asbestos, once hailed for its heat resistance and strength, was later linked to serious health issues, such as lung cancer and mesothelioma [84]. The "Radium Girls" suffered grave health consequences from radium exposure, a material they were told was safe to put in their mouths [85]. Tobacco, initially marketed as a harmless pastime, was found to be a primary cause of lung cancer and other health problems [86]. CFCs, once considered harmless and used to manufacture aerosol sprays and refrigerants, were found to deplete the ozone layer [87]. Thalidomide, a drug intended to alleviate morning sickness in pregnant women, led to severe birth defects [88]. And more recently, the proliferation of social media has been linked to an increase in depression and anxiety, especially among young people [89].

This emphasizes the importance of not only conducting expert testing but also implementing slow rollouts of technologies, allowing the test of time to reveal and address potential flaws before they impact a larger population. Even in technologies adhering to rigorous safety and security standards, undiscovered vulnerabilities may persist, as demonstrated by the Heartbleed bug—a serious vulnerability in the popular OpenSSL cryptographic software library that remained undetected for years before its eventual discovery [90].

Furthermore, even state-of-the-art AI systems, which appear to have solved problems comprehensively, may harbor unexpected failure modes that can take years to uncover. For instance, while AlphaGo's groundbreaking success led many to believe that AIs had conquered the game of Go, a subsequent adversarial attack on another highly advanced Go-playing AI, KataGo, exposed a previously unknown flaw [91]. This vulnerability enabled human amateur players to consistently defeat the AI, despite its significant advantage over human competitors who are unaware of the flaw. More broadly, this example highlights that we must remain vigilant when dealing with AI systems, as seemingly airtight solutions may still contain undiscovered issues. In conclusion, accidents are unpredictable and hard to avoid, and understanding and managing potential risks requires a combination of proactive measures, slow technology rollouts, and the invaluable wisdom gained through steady time-testing.

4.2 Organizational Factors can Reduce the Chances of Catastrophe

Some organizations successfully avoid catastrophes while operating complex and hazardous systems such as nuclear reactors, aircraft carriers, and air traffic control systems [92, 93]. These organizations recognize that focusing solely on the hazards of the technology involved is insufficient; consideration must also be given to organizational factors that can contribute to accidents, including human factors, organizational procedures, and structure. These are especially important in the case of AI, where the underlying technology is not highly reliable and remains poorly understood.

Human factors such as safety culture are critical for avoiding AI catastrophes. One of the most important human factors for preventing catastrophes is safety culture [94, 95]. Developing a strong safety culture involves not only rules and procedures, but also the internalization of these practices by all members of an organization. A strong safety culture means that members of an organization view safety as a key objective rather than a constraint on their work. Organizations with strong safety cultures often exhibit traits such as leadership commitment to safety, heightened accountability where all individuals take personal responsibility for safety, and a culture of open communication in which potential risks and issues can be freely discussed without fear of retribution [96]. Organizations must also take measures to avoid alarm fatigue, whereby individuals become desensitized to safety concerns because of the frequency of potential failures. The Challenger Space Shuttle disaster demonstrated the dire consequences of ignoring these factors when a launch culture characterized by maintaining the pace of launches overtook safety considerations. Despite the absence of competitive pressure, the mission proceeded despite evidence of potentially fatal flaws, ultimately leading to the tragic accident [97].

Even in the most safety-critical contexts, in reality safety culture is often not ideal. Take for example, Bruce Blair, a former nuclear launch officer and senior fellow at the Brookings Institution. He once disclosed that before 1977, the US Air Force had astonishingly set the codes used to unlock intercontinental ballistic missiles to "00000000" [98]. Here, safety mechanisms such as locks can be rendered virtually useless by human factors.

A more dramatic example illustrates how researchers sometimes accept a non-negligible chance of causing extinction. Prior to the first nuclear weapon test, an eminent Manhattan Project scientist calculated the bomb could cause an existential catastrophe: the explosion might ignite the atmosphere and cover the Earth in flames. Although Oppenheimer believed the calculations were probably incorrect, he remained deeply concerned, and the team continued to scrutinize and debate the calculations right until the day of the detonation [99]. Such instances underscore the need for a robust safety culture.

A questioning attitude can help uncover potential flaws. Unexpected system behavior can create opportunities for accidents or exploitation. To counter this, organizations can foster a questioning attitude, where individuals continuously challenge current conditions and activities to identify discrepancies that might lead to errors or inappropriate actions [100]. This approach helps to encourage diversity of thought and intellectual curiosity, thus preventing potential pitfalls that arise from uniformity of thought and assumptions. The Chernobyl nuclear disaster illustrates the importance of a questioning attitude, as the safety measures in place failed to address the reactor design flaws and ill-prepared operating procedures. A questioning attitude of the safety of the reactor during a test operation might have prevented the explosion that resulted in deaths and illnesses of countless people.

A security mindset is crucial for avoiding worst-case scenarios. A security mindset, widely valued among computer security professionals, is also applicable to organizations developing AIs. It goes beyond a questioning attitude by adopting the perspective of an attacker and by considering worst-case, not just average-case, scenarios. This mindset requires vigilance in identifying vulnerabilities that may otherwise go unnoticed and involves considering how systems might be deliberately made to fail, rather than only focusing on making them work. It reminds us not to assume a system is safe simply because no potential hazards come to mind after a brief brainstorming session. Cultivating and applying a security mindset demands time and serious effort, as failure modes can often be surprising and unintuitive. Furthermore, the security mindset

emphasizes the importance of being attentive to seemingly benign issues or "harmless errors," which can lead to catastrophic outcomes either due to clever adversaries or correlated failures [101]. This awareness of potential threats aligns with Murphy's law—"Anything that can go wrong will go wrong"—recognizing that this can be a reality due to adversaries and unforeseen events.

Organizations with a strong safety culture can successfully avoid catastrophes. High Reliability Organizations (HROs) are organizations that consistently maintain a heightened level of safety and reliability in complex, high-risk environments [92]. A key characteristic of HROs is their preoccupation with failure, which requires considering worst-case scenarios and potential risks, even if they seem unlikely. These organizations are acutely aware that new, previously unobserved failure modes may exist, and they diligently study all known failures, anomalies, and near misses to learn from them. HROs encourage reporting all mistakes and anomalies to maintain vigilance in uncovering problems. They engage in regular horizon scanning to identify potential risk scenarios and assess their likelihood before they occur. By practicing surprise management, HROs develop the skills needed to respond quickly and effectively when unexpected situations arise, further enhancing an organization's ability to prevent catastrophes. This combination of critical thinking, preparedness planning, and continuous learning could help organizations to be better equipped to address potential AI catastrophes. However, the practices of HROs are not a panacea. It is crucial for organizations to evolve their safety practices to effectively address the novel risks posed by AI accidents above and beyond HRO best practices.

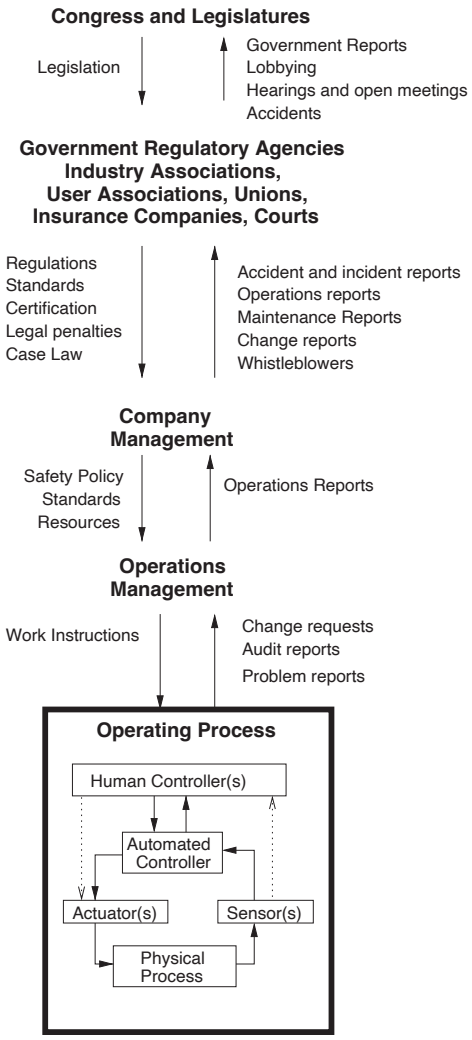

Figure 13: Mitigating risk requires addressing the broader sociotechnical system, including corporations (adapted from [94]).

Most AI researchers do not understand how to reduce overall risk from AIs. In most organizations building cutting-edge AI systems, there is often a limited understanding of what constitutes technical safety research. This is understandable because an AI's safety and intelligence are intertwined, and intelligence can help or harm safety. More intelligent AI systems could be more reliable and avoid failures, but they could also pose heightened risks of malicious use and loss of control. General capabilities improvements can improve aspects of safety, and it can hasten the onset of existential risks. Intelligence is a double-edged sword [102].

Interventions specifically designed to improve safety may also accidentally increase overall risks. For example, a common practice in organizations building advanced AIs is to fine-tune them to satisfy user preferences. This makes the AIs less prone to generating toxic language, which is a common safety metric. However, users also tend to prefer smarter assistants, so this process also improves the general capabilities of AIs, such as their ability to classify, estimate, reason, plan, write code, and so on. These more powerful AIs are indeed more helpful to users, but also far more dangerous. Thus, it is not enough to perform AI research that helps improve a safety metric or achieve a specific safety goal—AI safety research needs to improve safety *relative* to general capabilities.